Double-Angle and Half-Angle Formulas (6.4) Name ______________________

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write "2" in the blank. Solve that problem, find your answer and write "3" in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $\sin^2\theta$ # _______ If $\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$, find $\sin 2\theta$.	ANSWER: $\frac{\sqrt{2}}{2}$ # _______ Simplify $2\cos^2\theta - 1$.
ANSWER: $\frac{7}{25}$ # _______ Simplify $2\sin 15° \cos 15°$.	ANSWER: 1 # _______ If θ is in Quadrant I, is $\sin\frac{\theta}{2}$ positive or negative?

ANSWER: $\frac{24}{25}$ # ________ If $\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$, find $\cos 2\theta$.	ANSWER: $-\frac{1}{2}$ # ________ If $\sin\theta = \frac{\sqrt{2}}{2}$ and $\cos\theta = \frac{\sqrt{2}}{2}$, find $\sin 2\theta$.
ANSWER: $\frac{\sqrt{3}}{2}$ # ________ Simplify $\cos^2 22.5° - \sin^2 22.5°$.	ANSWER: positive # ________ Simplify $\frac{1-\cos 2\theta}{2}$.
ANSWER: $\frac{1}{2}$ # ________ Simplify $1 - 2\sin^2 15°$.	ANSWER: $\cos 2\theta$ # ________ If $\cos\theta = \frac{1}{2}$, find $\cos 2\theta$.